

Data Analytics - Project 1

Data Cleaning & Preparation Submission Report

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1. Objective

The goal of this project was to clean the raw dataset by identifying missing values, removing duplicates, correcting data formats, and preparing a reliable final dataset for analysis.

2. Dataset Overview

Metric	Value
Total rows in raw dataset	1200 records
Total columns	14 columns
Unique identifier	OrderID
Main cleaning columns	Date, CouponCode, Quantity, UnitPrice, TotalPrice, text fields

3. Cleaning Work Completed

ID	Action	Impact	Status
C001	Imputed missing CouponCode values	309 blank/null values replaced with "No Coupon".	Completed
C002	Corrected Date format	Dates converted from Excel serial display to yyyy-mm-dd format.	Completed
C003	Standardized numeric fields	Quantity and ItemsInCart as integers; UnitPrice and TotalPrice as 2-decimal values.	Completed
C004	Duplicate audit	0 duplicate OrderIDs and 0 exact duplicate rows found. No records removed.	Completed
C005	Validated TotalPrice	0 mismatches after checking TotalPrice = Quantity * UnitPrice.	Completed
C006	Trimmed text fields	Text values cleaned to support accurate grouping and filtering.	Completed

4. Final Validation Summary

Quality Check	Before Cleaning	After Cleaning	Result
Total records	1200	1200	Pass
Missing CouponCode values	309	0	Pass
Incorrectly formatted dates	Serial number display	0	Pass
Duplicate OrderIDs	0	0	Pass

Exact duplicate rows	0	0	Pass
TotalPrice mismatches	0	0	Pass

5. Submission Files

The Excel workbook contains five sheets: README, Raw_Data, Cleaned_Data, Validation_Summary, and Change_Log. The Cleaned_Data sheet is the final dataset for submission, while the Change_Log and Validation_Summary sheets document the audit trail and prove the quality gate was met.

Conclusion: The dataset is analysis-ready with 1200 records, 0 missing CouponCode values, 0 duplicate OrderIDs, 0 incorrect date formats, and 0 TotalPrice mismatches.